ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING INTERNSHIP

The internship report submitted in partial fulfillment of the requirements for the award of degree

BACHELOROF TECHNOLOGY IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JNTUA COLLEGE OF ENGINEERING (Autonomous), PULIVENDULA

ANDHRA PRADESH - INDIA

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Internship report on **Artificial Intelligence and Machine Learning Internship** is a Bonafide record of internship work done and submitted by **K.Nani,** bearing Roll No. **22191A0558**, in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

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With gratitude,
K.Nani
(**22191A0558**)

Internship Weekly Report Week – 1

(From 28/12/2025 To 03/01/2026)

Objective of the Activity Done: To set up the development environment and gain a strong foundation in web technologies, Django framework, and project planning for the Web Application Platform for Freelancers and Recruiters

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Project overview, requirement analysis, and understanding the platform scope for freelancers and recruiters.	Gained clarity on project goals, target users, and key features to be developed.
Day 2	Installed Python, Django, VS Code, and configured the development environment and virtual environment.	Successfully set up a working development environment for Django-based web development.
Day 3	Studied Django project structure, apps, settings, URLs, and the MTV (Model-Template-View) architecture.	Understood how Django organizes code and how requests flow through the framework.
Day 4	Designed the database schema — defined models for User, FreelancerProfile, RecruiterProfile, and JobListing.	Learned how to plan and design relational database models for a multi-role web platform.
Day 5	Applied Django migrations to create database tables and tested basic model operations.	Understood how Django ORM maps Python models to SQLite database tables.
Day 6	Explored Django Admin panel and registered models for easy data management and testing.	Learned how to use the Django Admin interface for project management and data inspection.

Internship Weekly Report Week – 2

(From 04/01/2026 To 10/01/2026)

Objective of the Activity Done: To implement user authentication, role-based access control, and user registration and login functionality for both freelancers and recruiters.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Implemented Django's built-in authentication system — user registration and login views.	Learned how to use Django's auth module for secure user management.
Day 2	Created separate registration forms for Freelancers and Recruiters with role-based user types.	Understood how to extend Django's User model and handle different user roles.
Day 3	Implemented login, logout, and session management with redirect logic based on user role.	Gained experience in managing user sessions and role-based redirection after login.
Day 4	Added form validation and error handling for registration and login forms.	Learned how to provide user-friendly feedback and validate form inputs in Django.
Day 5	Applied Bootstrap for responsive login and registration page design.	Improved UI consistency and responsiveness using Bootstrap components.
Day 6	Tested authentication flow end-to-end for both freelancer and recruiter roles.	Verified that authentication and role-based access were working correctly and securely.

Internship Weekly Report Week – 3

(From 11/01/2026 To 17/01/2026)

Objective of the Activity Done: To develop freelancer profile management features including profile creation, editing, resume upload, and profile image handling.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed the FreelancerProfile model with fields for skills, experience, bio, and contact details.	Learned how to design a comprehensive profile model for multi-field user data.
Day 2	Created profile creation and update views and forms using Django ModelForms.	Gained experience in building data-driven forms linked to database models.
Day 3	Implemented resume upload functionality using Django's FileField and MEDIA_ROOT settings.	Learned how to handle file uploads and manage media files in Django.
Day 4	Added profile image upload with image validation and storage configuration.	Understood how to manage image uploads and configure media file paths.
Day 5	Built the freelancer profile view page and designed the profile display template.	Gained skills in rendering dynamic model data in HTML templates.
Day 6	Tested profile creation, updating, and file upload features with different user accounts.	Verified that profile management features were working reliably across multiple users.

Internship Weekly Report Week – 4

(From 18/01/2026 To 24/01/2026)

Objective of the Activity Done: To develop the job listing module allowing recruiters to post, edit, and manage job opportunities on the platform.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed the JobListing model with fields for title, description, budget, deadline, and recruiter reference.	Learned how to model job listings with relational links to the recruiter's profile.
Day 2	Built job creation and editing forms and views for recruiters using Django class-based views.	Gained experience in using Django class-based views for CRUD operations.
Day 3	Implemented job listing display pages with filtering by category, skill, and status.	Understood how to add query-based filtering to Django views and templates.
Day 4	Added job deletion with confirmation prompts and permission checks (only recruiter can delete own jobs).	Learned how to implement access control and confirmation workflows in Django.
Day 5	Designed a recruiter dashboard showing all posted jobs with status indicators.	Gained skills in building role-specific dashboards with dynamic data rendering.
Day 6	Tested the entire job posting flow — creating, editing, and deleting listings as a recruiter.	Confirmed that all job listing features were functioning correctly with appropriate access controls.

Internship Weekly Report Week – 5

(From 25/01/2026 To 31/01/2026)

Objective of the Activity Done: To implement the job application module enabling freelancers to browse job listings, apply for jobs, and track their applications.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed the JobApplication model with fields for applicant, job, cover letter, status, and timestamp.	Learned how to model many-to-many relationships between freelancers and job listings.
Day 2	Built the job application submission view and form for freelancers.	Understood how to capture and save user-submitted applications to the database.
Day 3	Implemented application status tracking — Pending, Accepted, Rejected — visible to both freelancers and recruiters.	Learned how to manage dynamic status updates in a relational database model.
Day 4	Built the freelancer dashboard showing all submitted applications with their current status.	Gained skills in building personalized dashboards with filtered data for specific users.
Day 5	Implemented recruiter view to see all applications for each job listing and manage their status.	Understood how to build a recruiter-side management interface for reviewing applications.
Day 6	Tested the application flow end-to-end — submission, status update, and dashboard display.	Verified that the application module was functioning correctly for both user roles.

Internship Weekly Report Week – 6

(From 01/02/2026 To 07/02/2026)

Objective of the Activity Done: To enhance the platform with project management features, including project assignment, progress tracking, and communication tools.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed the Project model with fields for title, assigned freelancer, recruiter, deadline, and status.	Learned how to model project entities with multi-role relationships in the database.
Day 2	Built project creation views for recruiters to assign projects to selected freelancers.	Gained experience in building assignment-based workflow features in Django.
Day 3	Implemented project status tracking — Not Started, In Progress, Completed — with update controls.	Understood how to build status management workflows for project lifecycle tracking.
Day 4	Designed project detail pages visible to both freelancer and recruiter with timeline and status info.	Learned how to render detailed project data with appropriate role-based visibility.
Day 5	Added messaging/comment feature within a project for communication between freelancer and recruiter.	Gained skills in implementing simple in-project messaging using Django models.
Day 6	Tested project creation, assignment, status update, and messaging features.	Verified that project management features were working accurately for both user roles.

Internship Weekly Report Week – 7

(From 08/02/2026 To 14/02/2026)

Objective of the Activity Done: To implement search and filtering functionality across job listings and freelancer profiles to improve platform usability.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed search logic for job listings using Django ORM Q objects to support keyword-based search.	Learned how to implement dynamic search queries using Django's ORM Q object.
Day 2	Added filter options for job listings by skill category, budget range, and posting date.	Understood how to combine multiple filter conditions in Django views for refined search results.
Day 3	Implemented freelancer profile search for recruiters to find freelancers by skill or experience.	Gained experience in building role-specific search features targeting different data models.
Day 4	Integrated search forms into the job listing and freelancer listing pages with GET request handling.	Learned how to process search form inputs via GET requests and pass results to templates.
Day 5	Improved search UI with Bootstrap-styled search bars, filter dropdowns, and result count display.	Enhanced platform usability by improving the visual design of search and filter components.
Day 6	Tested search and filter functionality with various inputs and edge cases.	Confirmed that search and filtering returned accurate, relevant results under different conditions.

Internship Weekly Report Week – 8

(From 15/02/2026 To 21/02/2026)

Objective of the Activity Done: To develop the REST API layer using Django REST Framework (DRF) for enabling data exchange between the backend and frontend.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Installed Django REST Framework and configured it in the project settings.	Learned how to integrate DRF into an existing Django project for API development.
Day 2	Built serializers for JobListing, FreelancerProfile, and JobApplication models.	Understood how to convert complex Django model data into JSON format using serializers.
Day 3	Created RESTful API endpoints for listing, creating, and retrieving job listings.	Gained experience in building standard REST API endpoints using Django REST Framework views.
Day 4	Implemented API endpoints for freelancer profile retrieval and job application submission.	Learned how to expose database model data via JSON APIs for frontend consumption.
Day 5	Added token-based authentication to secure the REST API endpoints.	Understood how to protect API endpoints using DRF's token authentication mechanism.
Day 6	Tested all API endpoints using Postman — verified responses, status codes, and error handling.	Confirmed that all APIs were returning correct data and handling errors appropriately.

Internship Weekly Report Week – 9

(From 22/02/2026 To 28/02/2026)

Objective of the Activity Done: To improve the frontend UI using Bootstrap, enhance responsiveness, and develop a consistent visual design across all platform pages.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Audited all existing pages for UI inconsistencies and planned a unified Bootstrap-based design system.	Understood the importance of design consistency and how to plan a UI improvement strategy.
Day 2	Redesigned the homepage with a hero section, feature highlights, and call-to-action buttons.	Learned how to build an engaging landing page using Bootstrap grid and utility classes.
Day 3	Improved the job listing page layout with card-based job tiles and badge indicators for status.	Gained experience in using Bootstrap cards and badges for structured content display.
Day 4	Redesigned the freelancer profile page with a clean layout for skills, experience, and portfolio.	Understood how to present profile data in a visually organized and readable format.
Day 5	Applied responsive navbar, sidebar, and footer across all pages for consistent navigation.	Learned how to implement consistent navigation components across a multi-page web application.
Day 6	Tested responsiveness across different screen sizes using browser developer tools.	Verified that all pages displayed correctly on mobile, tablet, and desktop screen sizes.

Internship Weekly Report Week – 10

(From 01/03/2026 To 07/03/2026)

Objective of the Activity Done: To implement notification features and improve user communication by adding alerts, email notifications, and in-platform notification displays.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed the Notification model to store platform alerts for users (application updates, project assignments).	Learned how to model notification data and link it to specific users and events.
Day 2	Implemented notification creation triggers — auto-generate notifications on key events (application status change, project assignment).	Understood how to use Django signals to trigger automated actions on model events.
Day 3	Built the notification display view and notification bell icon in the navbar with unread count badge.	Gained skills in building real-time-style notification UIs using server-side rendering.
Day 4	Added mark-as-read functionality for individual and all notifications.	Learned how to manage notification state using simple database flag updates.
Day 5	Configured Django email backend and implemented email notifications for application status updates.	Understood how to send transactional emails from Django using the email backend.
Day 6	Tested notification creation, display, read status, and email delivery across both user roles.	Confirmed that notifications were delivered accurately and email alerts were functional.

Internship Weekly Report Week – 11

(From 08/03/2026 To 14/03/2026)

Objective of the Activity Done: To perform thorough debugging, bug fixing, and code refactoring to improve platform stability, code quality, and performance.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Identified and fixed bugs in the job application module — duplicate applications and status update errors.	Learned systematic debugging techniques and how to trace issues through Django logs.
Day 2	Resolved media file serving issues for resumes and profile images in development and production settings.	Understood Django's MEDIA_URL and MEDIA_ROOT configuration for file serving.
Day 3	Refactored views to reduce code duplication — extracted common logic into utility functions and mixins.	Gained experience in writing cleaner, reusable Django code using mixins and helper functions.
Day 4	Improved database query performance using select_related() and prefetch_related() for related models.	Learned how to optimize Django ORM queries to reduce database load and improve speed.
Day 5	Fixed form validation edge cases and improved error messaging across all forms.	Enhanced user experience by ensuring all forms provide clear, accurate validation feedback.
Day 6	Conducted a full platform walkthrough to identify remaining issues and verified all core features.	Gained confidence in the platform's stability and confirmed all major features were working correctly.

Internship Weekly Report Week – 12

(From 15/03/2026 To 21/03/2026)

Objective of the Activity Done: To implement security enhancements including CSRF protection, access control enforcement, input sanitization, and secure file upload handling.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Reviewed Django's built-in CSRF protection and verified all forms included CSRF tokens.	Understood how CSRF attacks work and how Django prevents them by default.
Day 2	Implemented login-required decorators and permission checks on all sensitive views.	Learned how to enforce access control and prevent unauthorized access to protected pages.
Day 3	Added object-level permission checks — ensured users can only edit or delete their own data.	Gained experience in implementing fine-grained object-level authorization in Django.
Day 4	Implemented file type validation for resume and image uploads to prevent malicious file uploads.	Understood how to validate uploaded file types and sizes to improve platform security.
Day 5	Applied input sanitization and escaping in templates to prevent XSS vulnerabilities.	Learned how Django's auto-escaping protects against cross-site scripting attacks.
Day 6	Conducted a security review of all views, forms, and file handling — documented findings.	Identified and resolved key security gaps, improving overall platform security posture.

Internship Weekly Report Week – 13

(From 22/03/2026 To 28/03/2026)

Objective of the Activity Done: To perform comprehensive testing including unit testing, integration testing, and user acceptance testing across all platform modules.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Wrote unit tests for Django models — tested field validations, relationships, and custom methods.	Learned how to write effective unit tests for Django models using the TestCase class.
Day 2	Wrote unit tests for views — tested response codes, redirects, and template rendering.	Gained experience in testing Django views and verifying expected HTTP responses.
Day 3	Performed integration testing — tested complete user workflows from registration to job application.	Understood how integration tests validate the interaction between multiple system components.
Day 4	Tested REST API endpoints with different input scenarios and edge cases using Postman.	Verified that API endpoints handled valid and invalid inputs correctly with proper error responses.
Day 5	Conducted user acceptance testing (UAT) — simulated real user interactions for both freelancer and recruiter flows.	Gained practical experience in validating platform usability from a real user's perspective.
Day 6	Documented all test cases and test results — created a test report summarizing coverage and outcomes.	Learned the importance of test documentation and how to report testing results professionally.

Internship Weekly Report Week – 14

(From 29/03/2026 To 04/04/2026)

Objective of the Activity Done: To deploy the web application to a production-ready environment and configure GitHub for version control and collaborative code management.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Organized the project directory structure and cleaned up unused files and debug settings.	Learned how to prepare a Django project for production deployment.
Day 2	Configured Django production settings — DEBUG=False, ALLOWED_HOSTS, static files, and secret key management.	Understood how to securely configure a Django application for a production environment.
Day 3	Set up a GitHub repository, committed the full project, and organized branches for development and main.	Gained experience in using GitHub for version control and managing code with branches.
Day 4	Collected and configured static files using collectstatic and configured media file serving.	Learned how to handle static and media file configuration for Django production deployments.
Day 5	Tested the production configuration locally — verified all features working under production settings.	Confirmed that the platform was stable and fully functional under production-like settings.
Day 6	Documented the deployment steps, project structure, and environment setup in a README file.	Understood the importance of documentation for project handover and future maintenance.

Internship Weekly Report Week – 15

(From 05/04/2026 To 11/04/2026)

Objective of the Activity Done: To prepare the internship report, project documentation, and presentation materials summarizing all work completed during the internship.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Outlined the internship report structure — introduction, weekly reports, project documentation, and conclusion.	Learned how to plan and structure a professional technical internship report.
Day 2	Wrote the project documentation — system architecture, database schema, and module descriptions.	Gained experience in documenting a software project comprehensively and clearly.
Day 3	Prepared the PowerPoint presentation covering project overview, tech stack, features, and demo screenshots.	Understood how to communicate a technical project effectively through a structured presentation.
Day 4	Reviewed and refined the internship weekly reports for accuracy and completeness.	Developed attention to detail and professional writing skills for technical documentation.
Day 5	Practiced the project presentation — rehearsed demo walkthrough and technical explanation.	Built confidence in presenting technical work and answering questions about the project.
Day 6	Final review of all documentation — report, presentation, and code repository.	Ensured all internship deliverables were complete, accurate, and professionally presented.

Internship Weekly Report Week – 16

(From 12/04/2026 To 18/04/2026)

Objective of the Activity Done: To finalize the platform, implement last improvements based on review feedback, and prepare the complete internship submission package

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Incorporated mentor feedback — improved the platform UI and fixed minor bugs identified during review.	Learned how to receive feedback constructively and apply improvements effectively.
Day 2	Added final enhancements — improved job listing card layout and freelancer search result display.	Gained experience in iterative development and UI refinement based on user feedback.
Day 3	Conducted final end-to-end testing of all platform features — confirmed all modules working correctly.	Verified the platform's readiness for submission and demonstrated professional quality assurance.
Day 4	Finalized the GitHub repository — cleaned up commits, updated README, and organized project files.	Understood how to maintain a clean, well-documented code repository for professional submission.
Day 5	Compiled the final internship report with all chapters, weekly logs, and project documentation.	Completed a comprehensive, professionally formatted internship report for academic submission.
Day 6	Submitted the internship report, presentation, and GitHub repository link to the mentor for final evaluation.	Successfully completed all internship deliverables within the specified duration.

Internship Weekly Report Week – 17

(From 19/04/2026 To 28/04/2026)

Objective of the Activity Done: To complete the final submission, present the project to the department, and reflect on the learning outcomes of the entire internship.

DAY	BRIEF DESCRIPTION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Prepared the final project demonstration — organized demo script and screenshots for presentation.	Developed presentation planning skills and learned how to structure a technical demo effectively.
Day 2	Delivered the project presentation to the mentor and department — demonstrated platform features and explained architecture.	Built professional communication skills and confidence in presenting technical work to an audience.
Day 3	Responded to mentor's technical questions and addressed feedback on the project and report.	Learned how to defend technical decisions and communicate design choices clearly.
Day 4	Prepared the internship completion summary — documented key learnings, challenges, and outcomes.	Reflected on the internship journey and articulated technical and professional growth.
Day 5-6	Formal closure of internship — submitted all final documents and received confirmation of successful internship completion.	Successfully concluded the internship with all deliverables submitted and acknowledged by the mentor.

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1. INTRODUCTION

About The Company

Codec Technologies Pvt. Ltd. is a technology-driven organization focused on providing industry-oriented training and internship programs in emerging domains such as Artificial Intelligence, Machine Learning, Data Science, and Software Development. The company aims to bridge the gap between academic learning and industry requirements by offering hands-on training and real-time project experience.

Codec Technologies provides students with exposure to modern tools, frameworks, and methodologies used in the development of intelligent systems. Through structured training modules and guided mentorship, interns gain practical knowledge and develop problem-solving skills required for real-world applications.

During my internship at **Codec Technologies Pvt. Ltd.**, I worked as an **Artificial Intelligence and Machine Learning Intern**. The internship provided a well-structured learning path starting from Python programming, data preprocessing, and exploration data analysis to machine learning algorithms and deep learning techniques.

The internship also involved working on a real-time project titled **"Platform for Freelancers and Recruiters"**, which focused on applying deep learning techniques for image classification.

This internship experience has significantly enhanced my technical knowledge in AI/ML and strengthened my analytical thinking and problem-solving abilities. It served as a bridge between academic concepts and practical implementation, preparing me to contribute effectively in the field of Artificial Intelligence and Machine Learning.

2. ABOUT THE INTERNSHIP

The internship offered by **Codec Technologies Pvt. Ltd.** provided an immersive learning experience designed to bridge the gap between academic knowledge and industry-ready skills in the field of **Artificial Intelligence and Machine Learning**. The program focused on hands-on, project-based learning, enabling participants to work on real-world problems while gaining exposure to modern AI/ML technologies.

The internship followed a structured learning progression spanning from **28/12/2025 to 28/04/2026**. The technologies and concepts covered during the internship included Python programming, data preprocessing, exploratory data analysis, machine learning algorithms, deep learning techniques, and model evaluation methods. Interns also gained experience with industry-standard tools and frameworks such as **NumPy, Pandas, Scikit-learn, TensorFlow, and Keras**.

One of the key highlights of this internship was the focus on developing real-world AI solutions. As part of the internship, I worked on a major project titled **"Platform for Freelancers & Recruiters"**, which involved building an image classification model using deep learning techniques.

The internship emphasized the complete lifecycle of an AI/ML project, including data collection, preprocessing, model building, training, evaluation, optimization, and deployment using a Flask-based web application. This approach helped in developing strong analytical skills, problem-solving ability, and practical knowledge of AI system development.

Overall, this internship provided a strong foundation in Artificial Intelligence and Machine Learning and enhanced my ability to apply theoretical knowledge to real-world scenarios, making me better prepared for future roles in the AI/ML domain.

3. INTERNSHIP SPECIFICATION

What is an Artificial Intelligence G Machine Learning Intern

An **Artificial Intelligence and Machine Learning (AI/ML) Intern** is trained to develop intelligent systems that can analyze data, learn from patterns, and make predictions or decisions. AI/ML involves building models and algorithms that simulate human intelligence to solve complex real-world problems.

During this internship, the role involved working with data, developing machine learning models, and implementing deep learning techniques for real-world applications such as image classification and prediction systems.

Technologies Used

Python:
Python is the primary programming language used in AI/ML due to its simplicity and powerful ecosystem. It is used for data processing, model building, and implementation of algorithms.

NumPy & Pandas:
These libraries are used for data manipulation and analysis. NumPy helps in numerical computations, while Pandas is used for handling structured datasets.

Matplotlib & Seaborn:
These libraries are used for data visualization. They help in understanding patterns, trends, and relationships within the data.

Scikit-learn:
A machine learning library used for implementing algorithms such as Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors, and Support Vector Machines.

Django REST Framework (DRF)
DRF is used to build RESTful APIs that enable seamless data exchange between the Django backend and frontend components.

SQLite
SQLite is a lightweight, serverless database engine used to store all platform data locally, making it ideal for development and small-scale deployment.

Flask:
A lightweight web framework used to deploy machine learning models into a web application, allowing users to interact with the system.

4. INTERNSHIP REQUIREMENTS

Development Environment Setup

The following tools were installed and configured for the internship:

1. Python (3.8+): Primary programming language for Django-based web development.

2. Django (4.x): Web framework used for backend development, ORM, and admin management.

3. VS Code / PyCharm: Development environment for writing, running, and debugging Python/Django code.

4. SQLite: Lightweight database engine for storing platform data during development.

5. Django REST Framework: Used for building RESTful APIs for data exchange between backend and frontend.

6. Bootstrap (5.x): CSS framework used for responsive, mobile-friendly frontend design.

7. Git & GitHub: Version control for tracking progress and managing the codebase collaboratively.

8. Postman: Used for testing and verifying REST API endpoints.

Python Libraries Installed

The following Python libraries were installed using pip:

- **numpy** - for numerical operations
- **pandas** - for data analysis and manipulation
- **matplotlib** - for basic data visualization
- **seaborn** - for advanced visualization
- **scikit-learn** - for machine learning algorithms
- **tensorflow / keras** - for deep learning models
- **opencv-python** - for image processing

5. MODULES LEARNED

Overview of Artificial Intelligence & Machine Learning

The Artificial Intelligence and Machine Learning internship provided a comprehensive understanding of building intelligent systems using data-driven approaches. The program covered Python programming, data preprocessing, machine learning algorithms, deep learning techniques, and model deployment — progressing from basic concepts to a complete real-world AI project.

Purpose

The purpose of the internship was to provide structured exposure to the complete AI/ML workflow — from data collection and preprocessing to model building, evaluation, and deployment. The internship prepared the intern for real-world roles in Artificial Intelligence and Machine Learning.

Objectives

Skill Development:
Gain practical experience in Python programming, data analysis, machine learning, and deep learning techniques.

Understanding AI/ML Workflow:
Learn how data is processed, models are trained, evaluated, and deployed in real-world applications.

Module-Based Learning:
Cover structured modules including Python basics, data preprocessing, machine learning algorithms, deep learning, and transfer learning.

Professional Development:
Develop analytical thinking, problem-solving skills, and project implementation ability.

Career Preparation:
Equip the intern for entry-level roles such as AI/ML Engineer, Data Analyst, and Machine Learning Intern through hands-on project experience.

6. ANALYSIS OF THE MODULES COMPLETED AS A PART OF INTERNSHIP

The internship provided a structured learning journey from **28/12/2025 to 28/04/2026**, progressing through Python programming, data preprocessing, machine learning algorithms, deep learning, and finally a real-world AI project using transfer learning. Below is a detailed analysis of each module.

Project Setup & Django Foundations (Week 1)

Content Covered: Django project structure, models, migrations, ORM basics, and admin panel configuration.

Practical Applications: Set up the development environment, designed the database schema, and configured the Django admin interface.

Analysis: Established the core project foundation required for all subsequent feature development.

User Authentication & Role-Based Access (Week 2)

Content Covered: Django authentication, user registration, login, session management, and role-based access control.

Practical Applications: Built separate registration and login flows for freelancers and recruiters with role-based redirection.

Analysis: Delivered a secure, role-aware authentication system that forms the backbone of the platform's access model.

Profile Management & File Handling (Week 3)

Content Covered: Django ModelForms, FileField, MEDIA_ROOT configuration, and profile image management.

Practical Applications: Developed freelancer profile creation, editing, resume upload, and profile image handling features.

Analysis: Addressed the real-world challenge of securely handling user-uploaded files in a Django web application.

Job Listing Module (Week 4)

Content Covered: Django class-based views, CRUD operations, filtering, and recruiter-specific access control.

Practical Applications: Built a complete job posting system for recruiters with listing management and status indicators.

Analysis: Delivered the core job-side functionality of the platform with proper access control enforcement.

Job Application Module (Week 5)

Content Covered: Application models, many-to-many relationships, status tracking, and personalized dashboards.

Practical Applications: Implemented the full application workflow — submission, status management, and role-specific dashboards.

Analysis: Connected the two sides of the platform — enabling freelancers to apply and

recruiters to manage applications.

Project Management & Communication (Week 6)

Content Covered: Project assignment models, status lifecycle, in-project messaging, and role-based project views.

Practical Applications: Developed project assignment, progress tracking, and communication features for active projects.

Analysis: Delivered the collaboration layer that enables effective project delivery between freelancers and recruiters.

Search & Filtering (Week 7)

Content Covered: Django ORM Q objects, multi-criteria filtering, GET-based search, and Bootstrap search UI.

Practical Applications: Built keyword search for jobs and freelancer profiles with category, skill, and date filters.

Analysis: Significantly improved platform usability by enabling efficient discovery of jobs and talent.

REST API Development (Week 8)

Content Covered: Django REST Framework, serializers, API views, and token-based authentication.

Practical Applications: Built REST APIs for job listings, profiles, and applications with secure token-based access.

Analysis: Enabled structured data exchange and prepared the platform for potential mobile or third-party integrations.

UI Enhancement & Responsiveness (Week 9)

Content Covered: Bootstrap 5 components, responsive design, card layouts, and consistent navigation.

Practical Applications: Redesigned all pages with a consistent, responsive Bootstrap-based layout and improved visual design.

Analysis: Elevated the platform's professional appearance and ensured a quality user experience across all devices.

Notifications & Email Alerts (Week 10)

Content Covered: Django signals, notification models, email backend, and unread count badges.

Practical Applications: Implemented in-platform notifications and email alerts for application updates and project assignments.

Analysis: Improved user engagement and ensured both freelancers and recruiters stay informed about key events.

Debugging, Refactoring & Optimization (Week 11)

Content Covered: Bug fixing, ORM query optimization (select_related, prefetch_related), and code refactoring.

Practical Applications: Identified and fixed bugs, optimized database queries, and improved code structure across the project.

Analysis: Improved platform stability, performance, and code maintainability significantly.

Security Implementation (Week 12)

Content Covered: CSRF protection, login-required decorators, object-level permissions, and input sanitization.

Practical Applications: Enforced access control, validated file uploads, and secured all views against common web vulnerabilities.

Analysis: Delivered a security-hardened platform meeting standard web application security practices.

Testing (Week 13)

Content Covered: Unit testing, integration testing, API testing with Postman, and user acceptance testing.

Practical Applications: Wrote and executed test cases for models, views, APIs, and complete user workflows.

Analysis: Validated platform reliability and correctness across all modules through structured testing.

Deployment & Version Control (Week 14)

Content Covered: Django production settings, static/media file configuration, GitHub repository management.

Practical Applications: Prepared the project for production, configured GitHub, and documented deployment steps.

Analysis: Completed the full development lifecycle by successfully preparing and documenting the platform for deployment.

7.PROJECT WORK UNDERTAKEN

Introduction

Introduction

The project titled "Web Application Platform for Freelancers and Recruiters" is a web application built using the Django framework. The platform serves as a unified digital workspace designed to connect freelancers seeking work with recruiters looking for skilled talent — solving the real-world problem of fragmented, untrustworthy freelance job markets.

Project Overview

The platform provides the following key features:

1. User Registration & Role-Based Access: Separate registration and login flows for freelancers and recruiters with role-specific dashboards.
2. Freelancer Profile Management: Profile creation and editing with skills, experience, bio, resume upload, and profile image.
3. Job Listing Module: Recruiters can post, edit, and manage job listings with detailed descriptions and requirements.
4. Job Application System: Freelancers can browse and apply for jobs; application status (Pending / Accepted / Rejected) is visible to both parties.
5. Project Management: Recruiters assign projects to accepted freelancers; project progress is tracked through status updates.
6. In-Platform Messaging: Communication between freelancer and recruiter within an active project.
7. Search & Filtering: Keyword-based search for job listings and freelancer profiles with multi-criteria filtering.
8. REST API: JSON APIs for job listings, profiles, and applications built with Django REST Framework.
9. Notification System: In-platform notifications and email alerts for key events.
10. Security & Access Control: CSRF protection, login-required views, object-level permissions, and input validation.

Architecture

Frontend (Template Layer)

Built using Django templates with Bootstrap 5. Provides the user interface for registration, job listings, profile pages, applications, dashboards, and project management. Fully responsive across all screen sizes.

Backend (Django Application Layer)

Built using Django's MTV architecture. Manages business logic, authentication, database operations, file handling, REST APIs, notifications, and security enforcement.

Database Layer

SQLite database accessed via Django ORM. Stores all user data, profiles, job listings, applications, projects, messages, and notifications.

Technologies Used

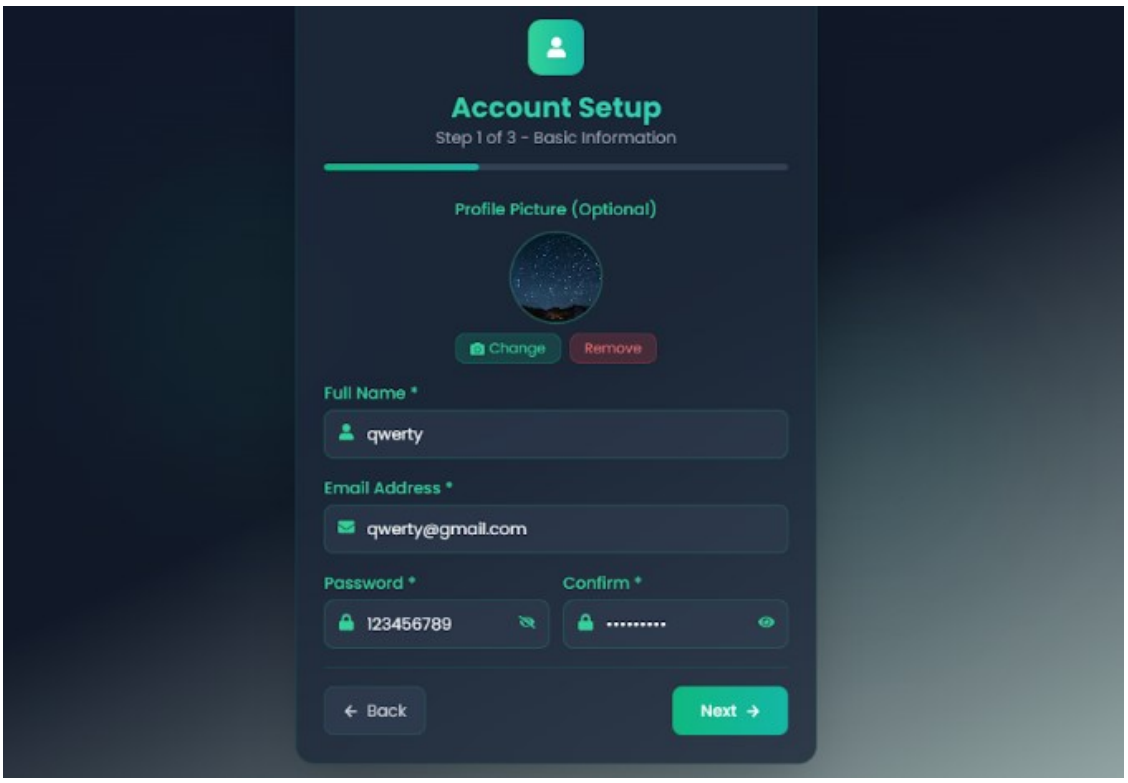
Technology	Purpose
Python & Django	Core backend framework for web application development
HTML, CSS, Bootstrap 5	Frontend design and responsive layout
JavaScript	Dynamic UI interactions and form validation
Django REST Framework	REST API development for data exchange
SQLite	Lightweight relational database for data storage
Django ORM	Object-relational mapping for database operations
Pillow	Image processing for profile image uploads
GitHub	Version control and code management
VS Code	Primary development environment
Postman	REST API testing and verification

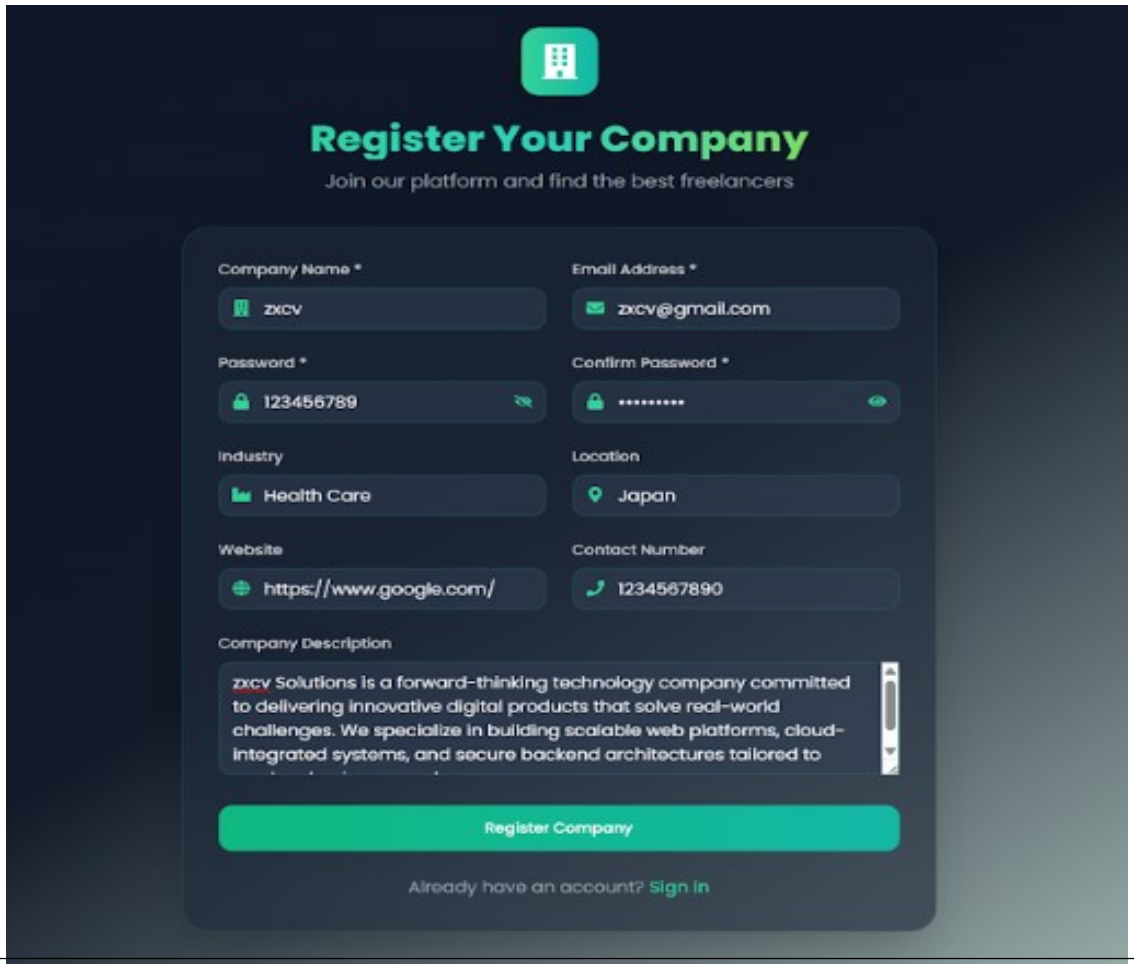
Testing

The platform was thoroughly tested to ensure functionality, security, and reliability:

- Unit tests for models and views using Django's TestCase framework.
- Integration testing of complete user workflows — registration through project.
- REST API testing using Postman with valid and invalid input scenarios.
- Security testing — verified CSRF, access control, and input sanitization.
- User acceptance testing simulating real freelancer and recruiter interactions.
- Responsive design testing across mobile, tablet, and desktop screen sizes.

Output Screenshots:





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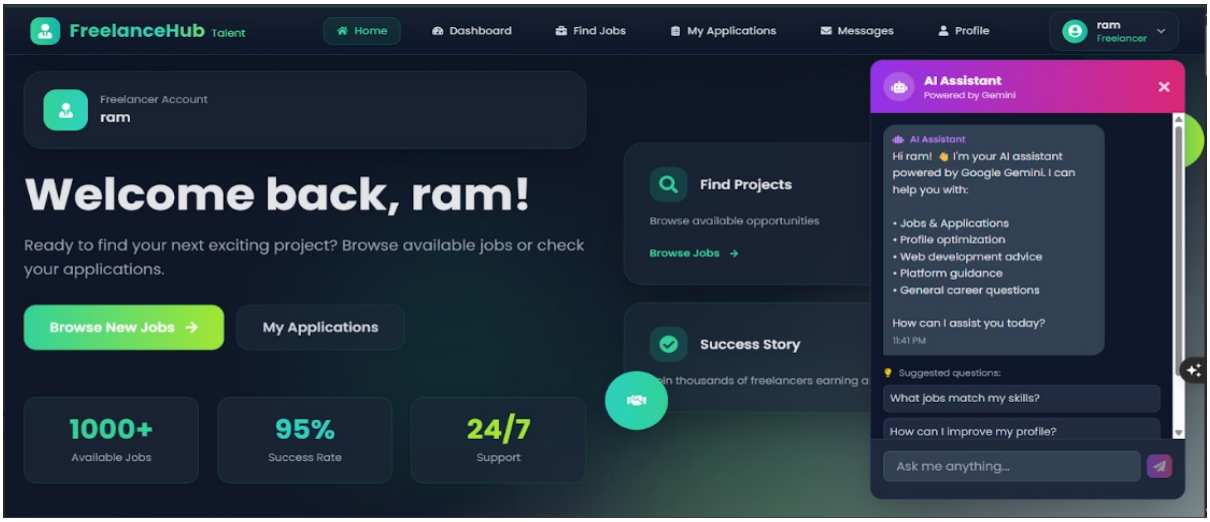
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Browse available opportunities
Browse Jobs
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AI Assistant
Hi ram! 👋 I'm your AI assistant powered by Google Gemini. I can help you with:
• Jobs & Applications
• Profile optimization
• Web development advice
• Platform guidance
• General career questions
How can I assist you today?
11:41 PM
Suggested questions:
What jobs match my skills?
How can I improve my profile?
Ask me anything...

8.CONCLUSION

This internship at **Codec Technologies Pvt. Ltd.** as an **Artificial Intelligence and Machine Learning Intern** provided an excellent learning experience spanning from **28/12/2025 to 28/04/2026**. The structured progression from Python programming and data preprocessing to machine learning and deep learning techniques equipped me with industry-relevant skills in the field of AI/ML.

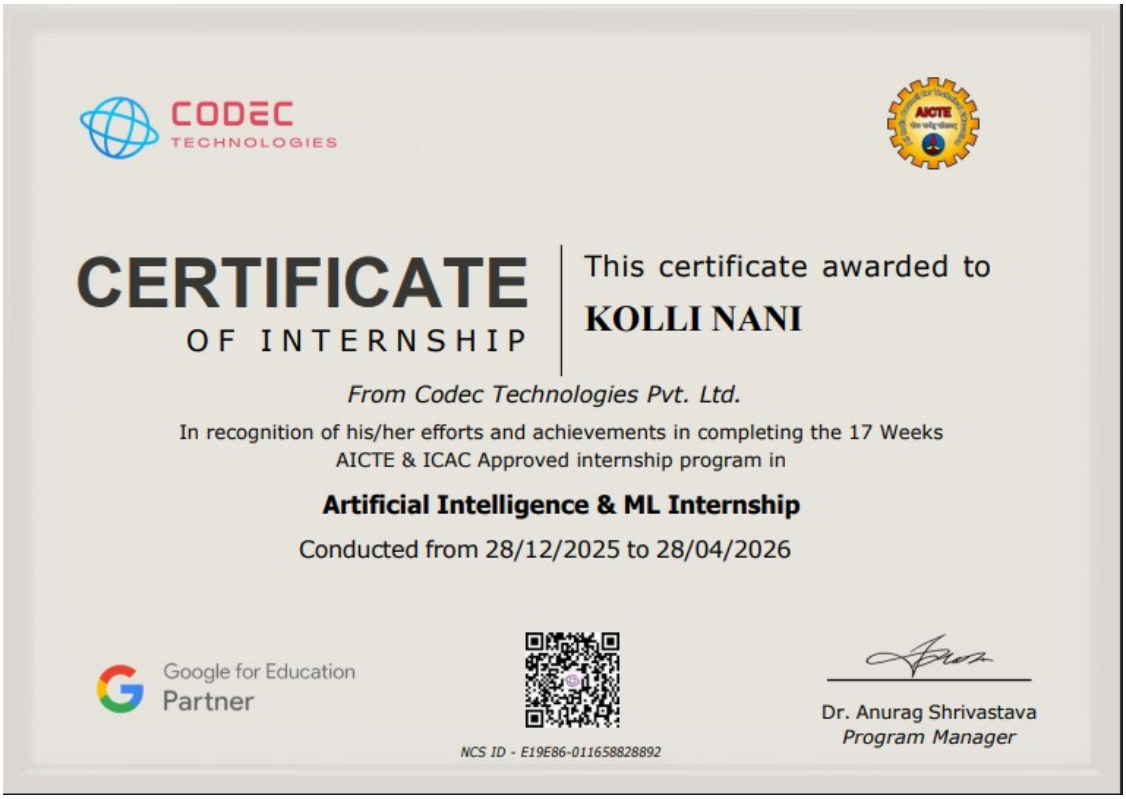
The project titled **"PLATFORM FOR FREELANCERS & RECRUITERS"** served as the culmination of all the concepts learned during the internship. Through this project, hands-on experience was gained in user authentication, role-based access control, full CRUD operations, REST API development, responsive UI design using Bootstrap, file handling, Django signals, security practices, and testing methodologies. The integration of all these concepts into a single deployable application provided a complete industry-level development experience.

This internship has significantly strengthened my analytical thinking, problem-solving ability, and technical skills in Artificial Intelligence and Machine Learning. It has prepared me to take on real-world challenges and pursue professional roles in AI/ML and data science domains.

9. INTERNSHIP CERTIFICATE

The internship certificate awarded by **Codec Technologies Pvt. Ltd.** is attached below as proof of successful completion of the **Artificial Intelligence and Machine Learning Internship**.

The certificate confirms that the internship was completed over a duration of **17 weeks (28/12/2025 to 28/04/2026)** and recognizes the efforts and achievements in the field of Artificial Intelligence and Machine Learning.



CODEC TECHNOLOGIES

CERTIFICATE OF INTERNSHIP

This certificate awarded to

KOLLI NANI

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 17 Weeks AICTE & ICAC Approved internship program in

Artificial Intelligence & ML Internship

Conducted from 28/12/2025 to 28/04/2026

Google for Education Partner

NCS ID - E19E86-011658828892

Dr. Anurag Shrivastava
Program Manager